

Manuscript_CofD_22.4.25.docx

by Mohammad Nayeem Hasan

Submission date: 22-Apr-2025 01:37AM (UTC-0700)

Submission ID: 2653302104

File name: Manuscript_CofD_22.4.25.docx (450.18K)

Word count: 4024

Character count: 24154

Trends and Disparities in Causes and Community-Based Disease Burden of Death in an Urban Bangladesh Community: A 2011-2021 Verbal Autopsy Study

Introduction:

Many low- and middle-income countries (LMICs) rely on hospital death records as their main source of causes of death (COD) information [1]. Hospital deaths, however, are only a subset of all deaths, mainly those that take place in medical facilities, therefore using them to estimate mortality rates for the wider population can introduce biases [2,3]. The COD distribution in the general population might not be accurately reflected by this. Skilled medical professionals use accounts of symptoms and situations to determine the cause. In places without civil registration systems, COD are often ascertained by verbal autopsy (VA) [4,5]. To monitor public health progress, comprehend epidemiological trends, and direct resource allocation, a functional vital registration (VR) system is necessary. While affluent nations usually have well-established VR systems that offer precise COD information, LMICs sometimes lack or have insufficient systems [6,7]. This disparity emphasizes how urgently these areas require accurate and representative statistics on the causes of death.

Mortality statistics by cause are limited in Bangladesh, and deaths that take place in private medical facilities or at home are sometimes not reported. Only one out of ten fatalities are formally documented, according to the Directorate General of Health Services (DGHS), and mostly morbidity codes rather than mortality codes from public hospitals are used in these records [8]. Health planning and resource distribution are made more difficult by this dearth of thorough data. Bangladesh lacks a thorough civil death registration system, just like many other developing nations. Therefore, routine, representative statistics on causes of death are needed to ensure that health initiatives and policies are effective [9]. A feasible and affordable way to produce population-based cause of death data is through VA, particularly in nations with insufficient medical certification of death and civil registration [10,11]. VA can offer important perspectives for developing policies and health planning.

It takes a significant financial commitment to set up a national death registration system in an area with inadequate resources. As a solution to this problem, we suggest a cemetery-based strategy that can be a practical and affordable way to register and notify people of deaths. Public cemeteries serve as important sites for death monitoring since they are frequently used for burial in metropolitan settings where land is expensive and in short supply. Bangladesh is home to more than 50 million people across 328 municipalities. Approximately one-third of the nation's population might be reached with this cemetery-based strategy with little funding. This study intends to investigate the prevalence of deaths by age and sex, as well as the association between socioeconomic factors and cause-specific diseases at the community level. In order to determine new health priorities and evaluate the burden of chronic diseases, the project will gather information on preventable and modifiable causes of mortality. It will also improve knowledge of

health habits and environmental risk factors, which will be useful to policymakers, doctors, and health program evaluators. In addition to providing important information on the deceased's health practices and risk factors, the data collected from verbal autopsies will help improve future health interventions and allocate resources more effectively.

Methods:

Study setting: Jamalpur Pauroshova, is comprised of around 35,000 households with a population size of approximately 150,000. This is a Muslim dominated (approximately 90%) community where almost all deceased are buried in the public cemeteries [12]. All cemeteries (three muslim and one Christian) and cremation centre (Hindu) of Jamalpur Pauroshova will be under surveillance for capturing death cases for VA interview.

Instrument: A validated and standardized VA tool was employed in this investigation to calculate COD. The VA program collects data from a close caregiver regarding the deceased's lifestyle behaviors, medical records, signs and symptoms of their terminal illness, and other attributes. This investigation was carried out using the WHO-Standardized verbal autopsy (VA) technique. In a verbal autopsy, family members or caregivers are questioned regarding the deceased's symptoms, signs, lifestyle choices, and other traits prior to death, as well as the circumstances surrounding their passing.

Data collection and analysis: A special form was created for cemetery-based death notifications, and cemetery attendants will fill it out. Following appropriate training by a group of medical professionals, one field worker was hired locally to guarantee a shared cultural background with the local community for data collecting. gathered data that is used to determine the most likely causes of death, usually by having a group of doctors independently analyze the data and attempt to agree on a single cause. To guarantee a shared cultural background with the local community, field workers will be hired locally. Bengali will be used to translate the verbal autopsy questionnaire that was modified from a standardized WHO. The causes of death will be ascertained via the physician review technique. SmartVA-Analyze software used to collect and analyse VA data. In our study, we classified the causes of death into seven distinct categories to facilitate analysis and interpretation. These included accidents, which encompassed unintentional injuries such as road traffic incidents, falls, and drowning. Cancer was categorized separately and included all forms of malignant tumors, such as those affecting the lungs, breast, or colon. Cardiovascular diseases referred to conditions like heart attacks and strokes, as well as other heart-related disorders. Additionally, we found a category called chronic diseases, which included long-term non-communicable diseases like diabetes and chronic renal disease, but did not include cardiovascular or cancerous problems. Asthma and chronic respiratory disorders like chronic obstructive pulmonary disease (COPD) were classified in the category of lung diseases, which was thought to be distinct from cancer. The other category was used to group causes that did not fit neatly into the aforementioned classifications, such as infectious diseases or liver disorders.

Finally, the unknown category accounted for cases where the cause of death was either not reported or could not be determined any single cause of deaths. In addition, the study period was divided into two-time frames: 2011–2016 and 2017–2021. Seasons were classified into three categories: Cool, Hot, and Rainy. Age was grouped into four categories: 0–17 years, 18–44 years, 45–64 years, and 65 years or older. Sex was categorized as Male and Female. Descriptive and inferential statistics conducted to examine the study's hypotheses. These tests may include a Bar chart, pie chart, cross-tabs, and chi-square test to assess the association. A p-value below 0.05 is considered indicative of statistical significance.

Ethical Considerations: This study conducted at Jamalpur Pauroshova/municipality where Biomedical Research Foundation (BRF), a partner of this project, has already established a community-based research platform and networks. This study will capitalize on these resources and experiences obtained from that community.

Results:

10,480 deaths occurred in all months throughout the 11-year period. With 754 deaths, 2013 was the year with the fewest deaths overall, while 2021 had the most deaths (1,270). According to monthly data, April had the fewest fatalities overall (732 deaths), while January had the most deaths overall (1,062 deaths) of any year. Regarding annual patterns, the number of deaths increased gradually over time, beginning in 2011 with 815 deaths and continuing to rise until reaching a peak of 1,270 deaths in 2021 (Figure 1).

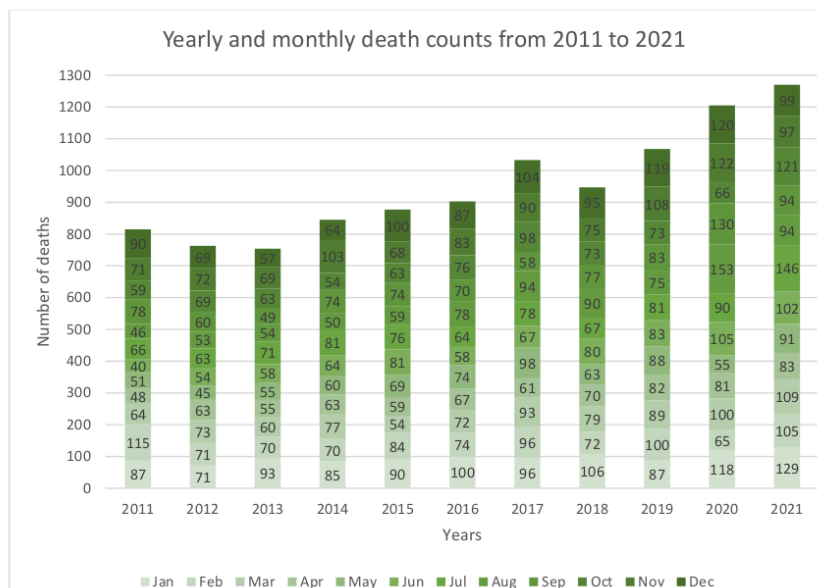


Figure 1: Monthly and yearly trends in the causes of death in an urban community in Bangladesh from 2011 to 2021

2

The top five causes of death in this town are unknown reasons, stroke, cancer, pneumonia, and other accidents. A total of 4,138 deaths were attributed to unknown causes, the highest number recorded, highlighting a significant portion of mortality that could not be classified. After this, stroke is one of the leading causes of mortality, accounting for 2,833 deaths, making it the most common cause. Other significant causes include cancer (537 deaths), pneumonia (346 fatalities), and other accidents (297 deaths) (Figure 2c).

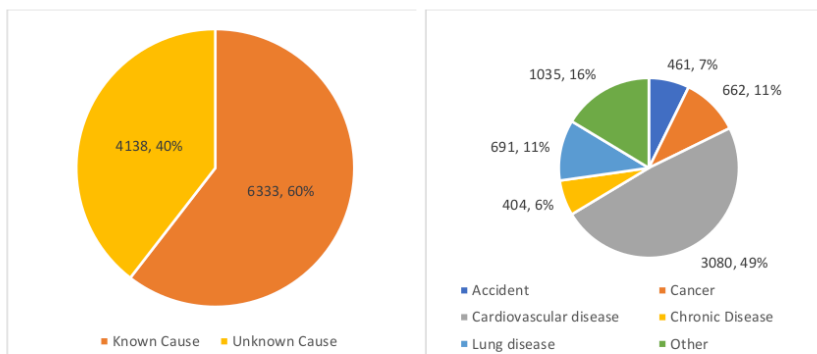


Figure 2a: Distribution of causes of death

Figure 2b: Types of death by known causes

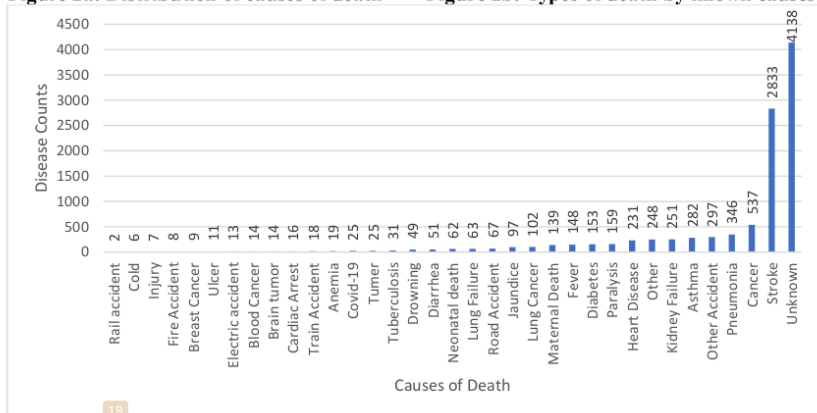


Figure 2c: Causes of Death in an Urban Community of Bangladesh (2011-2021)

Males accounted for 60.03% of deaths over the 2011–2021 timeframe, while females made up 39.97%. Male and female deaths were found to differ significantly by age group, with males dying more frequently in the 45–64 age range (64.89%) and females dying more frequently in the 65+ age range (43.32%). The causes of death for males and females differed significantly; accidents were more common in males (74.40%), whereas chronic diseases were more common in females (49.01%). The age group 65 and above had the largest percentage of deaths (47.66%), followed by the 45–64 age group (30.36%). Regarding seasonal variance, the cool season accounted for the largest percentage of deaths for those aged 65 and over (50.50%) and the lowest for those aged 0–17. While cardiovascular illnesses accounted for 29.42% of fatalities, with a notably high

percentage (86.81%) among the elderly (65 or above), unknown reasons accounted for the largest percentage of deaths overall (39.52%) (Table 1).

Accident	343 (74.40)	118 (25.60)	<0.001	91 (19.74)	178 (38.61)	129 (27.98)	63 (13.67)	<0.001	461 (4.40)
Cancer	367 (55.44)	295 (44.56)		24 (3.63)	134 (20.24)	369 (55.74)	135 (20.39)		662 (6.32)
Cardiovascular disease	2160 (70.13)	920 (29.87)		39 (1.27)	517 (16.79)	1676 (54.42)	848 (27.53)		3080 (29.42)
Chronic disease	206 (50.99)	198 (49.01)		5 (1.24)	57 (14.11)	259 (64.11)	83 (20.54)		404 (3.86)
Lung disease	445 (64.49)	245 (35.51)		356 (51.59)	48 (6.95)	165 (23.91)	121 (17.54)		690 (6.59)
Other	599 (57.87)	436 (42.13)		339 (32.75)	286 (27.63)	261 (25.21)	149 (14.40)		1035 (9.89)
Unknown	2165 (52.32)	1973 (47.68)		151 (3.65)	75 (1.81)	320 (7.73)	3592 (86.81)		4138 (39.52)
Total	6285 (60.03)	4185 (39.97)		1005 (9.60)	1295 (12.37)	3179 (30.36)	4991 (47.66)		10480 (100.00)

Discussion:

The study's conclusions emphasize the importance of varied causes of death as well as the significant differences in health burdens from particular non-communicable diseases (NCDs), TB, and injuries among age and sex categories in Bangladeshi urban neighborhoods. These findings highlight the necessity of focused public health priorities and actions. Similar age and sex discrepancies were also seen in the distribution of disease risk variables within these metropolitan populations [13]. However, a lot of non-communicable diseases (NCDs) can be avoided by altering one's behavior. Currently, 20–25% of the world's illness burden is caused by lifestyle-related variables, and in low-income nations, this percentage is predicted to increase quickly [14]. Both regular physical activity and fruit and vegetable eating are very low in Bangladesh. Alcohol consumption is very uncommon, but substance addiction, excessive salt consumption, and tobacco use are all very common [13]. Our findings correspond with earlier research that identified non-communicable diseases (NCDs) as a prominent cause of adult death, including among women, in Bangladesh [15–17]. According to a study by Labrique et al., which involved a group of women from rural Bangladesh, non-communicable diseases (NCDs) were the primary cause of mortality for women who were of reproductive age, making up 48% of all fatalities. Infectious disorders (17%), traumas (9%), and pregnancy-related causes (22%), were next. Circulatory disorders accounted for 40% of NCD-related deaths, with cancer coming in second at 15% [18].

Drowning mortality is significantly influenced by age, with children under five years old having the highest rate of drowning deaths. In Bangladesh, drowning claims the lives of around 14,000 youngsters annually. Drowning is still the second most common cause of death for children under five, despite getting little public attention. This underscores a serious but little-known public health emergency in the nation [19]. Key risk factors for drowning include lack of maternal education, children under five, especially males, and situations where newborns are left unattended, which is in line with findings from other studies [20]. Bangladesh has achieved significant strides in the fight against infectious illnesses. The employment of technologies to stop dehydration-related deaths from watery diarrhea is one noteworthy development that has the potential to significantly lower mortality rates [21]. Overall, our findings are consistent with those reported in the Global Burden of Disease (GBD) study [22,23] and align with WHO estimates regarding the leading causes of death in children [24]. Our results, however, indicate a lesser contribution from dengue and diarrheal illnesses in comparison to earlier investigations. These disparities may be explained, in part, by time patterns in cause-specific mortality. Rehydration therapy was first used in 1979, and since then, the number of children worldwide who die from severe diarrhea has dropped by up to two-thirds [25].

As one of the world's most climate-vulnerable countries, Bangladesh faces frequent natural disasters, causing significant loss of life and infrastructure. Since 1990, infectious disease mortality has declined notably—diarrheal deaths dropped by 90%, though further reduction is needed. Pneumonia rates remain high, requiring zinc and vaccination programs, while tuberculosis must be controlled to prevent a rise in HIV/AIDS. The tetanus vaccine would have the biggest effect

among the vaccinations. Pregnant women who receive the tetanus vaccine could almost completely prevent neonatal tetanus deaths [26]. The measles vaccine, which is likewise extremely effective, would have a big impact. However, there are significant obstacles because to its high cost and the requirement for a dependable cold chain for storage and delivery [27]. Although exact statistics are scarce, childhood TB and whooping cough (pertussis) may be responsible for some deaths from fevers and respiratory illnesses. There are effective vaccines for both illnesses that could be used extensively [28,29].

24

Strengths and limitations:

The main strength of the current study is using 11 years of death information consists of more than 10,000 deaths case to explore causes of deaths. This study's strengths include its inclusive, community-based design using cemetery surveillance across all religious groups and a large sample size collected over 11 years. The use of a WHO-standardized verbal autopsy tool ensured reliable and comparable cause-of-death data. Stratification by age, sex, season, and year allowed for detailed analysis, while locally recruited field workers ensured cultural sensitivity. The application of both descriptive and inferential statistics added further rigor to the findings. This percentage of fatalities classified as 'Unknown' reflects a methodological issue in accurately assigning causes of death using the verbal autopsy (VA) method. About 40% of fatalities in this study were categorized as "Unknown," either because the examining physicians were unable to agree on a single cause of death or because the information obtained from VA made it impossible to pinpoint the cause. The cause-specific fatality rates presented here may change slightly if these patients could be properly diagnosed. equivalent or greater percentages in the "unknown" or "unclassified" categories have been recorded in earlier research, though, that used equivalent verbal autopsy techniques to determine the causes of death [30]. Variations in diagnosis algorithms and the biases they introduce may account for some of the discrepancies between our findings and those of the GBD study. According to the verbal autopsy (VA) technique, the majority of causes of death have unique symptom complexes that lay respondents can identify, recollect, and report. Causes of death can be categorized into helpful groups based on the information that has been reported. Both adult and child mortality have been confirmed using the VA technique, and the findings can be modified to take misclassification biases into consideration. It is troublesome, nevertheless, to assume that the same correction procedure will work well at different locations. We couldn't apply general bias corrections to our dataset because we didn't have a complete set of validation experiments.

Conclusion:

This study highlights the shifting mortality patterns in an urban Bangladeshi community, with a rising burden of non-communicable diseases and a significant number of deaths from unknown

causes. Age, sex, and seasonal variations in causes of death emphasize the need for targeted health interventions. Strengthening community-based surveillance and preventive measures, including vaccination and lifestyle changes, is essential to reduce preventable deaths and improve public health outcomes. It takes creative and culturally relevant methods of health promotion to address these risk factors. Together with palliative care, prevention can improve rural residents' quality of life, lower needless medical expenses and lost productivity, and boost local and national economies.

References:

1. Jha P. Reliable direct measurement of causes of death in low- and middle-income countries. *BMC Med* [Internet]. 2014 Feb 4 [cited 2025 Apr 11];12(1):1–10. Available from: <https://bmcmmedicine.biomedcentral.com/articles/10.1186/1741-7015-12-19>
2. Kristoffersen DT, Helgeland J, Clench-Aas J, Laake P, Veierod MB. Comparing hospital mortality - How to count does matter for patients hospitalized for acute myocardial infarction (AMI), stroke and hip fracture. *BMC Health Serv Res* [Internet]. 2012 Oct 22 [cited 2025 Apr 11];12(1):1–10. Available from: <https://bmchealthservres.biomedcentral.com/articles/10.1186/1472-6963-12-364>
3. Meacock R, Anselmi L, Kristensen SR, Doran T, Sutton M. Do variations in hospital admission rates bias comparisons of standardized hospital mortality rates? A population-based cohort study. *Soc Sci Med*. 2019 Aug 1;235:112409.
4. Hazard RH, Buddhika MPK, Hart JD, Chowdhury HR, Firth S, Joshi R, et al. Automated verbal autopsy: From research to routine use in civil registration and vital statistics systems. *BMC Med* [Internet]. 2020 Mar 9 [cited 2025 Apr 11];18(1):1–11. Available from: <https://bmcmmedicine.biomedcentral.com/articles/10.1186/s12916-020-01520-1>
5. de Savigny D, Riley I, Chandramohan D, Odhiambo F, Nichols E, Notzon S, et al. Integrating community-based verbal autopsy into civil registration and vital statistics (CRVS): system-level considerations. *Glob Health Action* [Internet]. 2017 [cited 2025 Apr 11];10(1):1272882. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5328373/>
6. Mikkelsen L, Phillips DE, Abouzahr C, Setel PW, De Savigny D, Lozano R, et al. A global assessment of civil registration and vital statistics systems: Monitoring data quality and progress. *The Lancet* [Internet]. 2015 Oct 3 [cited 2025 Apr 11];386(10001):1395–406. Available from: <https://www.thelancet.com/action/showFullText?pii=S0140673615601714>
7. AbouZahr C, De Savigny D, Mikkelsen L, Setel PW, Lozano R, Nichols E, et al. Civil registration and vital statistics: Progress in the data revolution for counting and accountability. *The Lancet* [Internet]. 2015 Oct 3 [cited 2025 Apr 11];386(10001):1373–85. Available from: <https://www.thelancet.com/action/showFullText?pii=S0140673615601738>
8. Hossain AT, Siddique AB, Jabeen S, Khan S, Haider MM, Ameen S, et al. Maternal mortality in Bangladesh: Who, when, why, and where? A national survey-based analysis. *J Glob Health* [Internet]. 2023 [cited 2025 Apr 11];13:07002. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10248997/>

9. Uddin M, Ashrafi SAA, Azad AK, Chowdhury A, Chowdhury HR, Riley ID, et al. Improving coverage of civil registration and vital statistics, Bangladesh. *Bull World Health Organ* [Internet]. 2019 Sep 1 [cited 2025 Apr 11];97(9):637. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6705501/>
10. Alyazidi F, Shakely D, Alyazidi FR, Petzold M, Hussain-Alkhateeb L. Validating the diagnostic accuracy of medical certification: a population-level comparison between verbal autopsy and Saudi medical records causes of death of deceased with type 2 diabetes. *Glob Health Action* [Internet]. 2024 Dec 31 [cited 2025 Apr 11];17(1). Available from: <https://www.tandfonline.com/doi/abs/10.1080/16549716.2024.2448382>
11. Hazard RH, Buddhika MPK, Hart JD, Chowdhury HR, Firth S, Joshi R, et al. Automated verbal autopsy: from research to routine use in civil registration and vital statistics systems. *BMC Med* [Internet]. 2020 Mar 9 [cited 2025 Apr 11];18(1):60. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7061477/>
12. BBS. POPULATION & HOUSING CENSUS 2011 [Internet]. 2011 [cited 2025 Apr 11]. Available from: https://bbs.portal.gov.bd/sites/default/files/files/bbs.portal.gov.bd/page/7b7b171a_731a_4854_8e0a_f8f7dede4a4a/PHC2011PreliminaryReport.pdf
13. Ahmed SM, Hadi A, Razzaque A, Ashraf A, Juvekar S, Ng N, et al. Clustering of chronic non-communicable disease risk factors among selected Asian populations: levels and determinants. *Glob Health Action* [Internet]. 2009 [cited 2025 Apr 7];2(1):68–75. Available from: <https://pubmed.ncbi.nlm.nih.gov/20027260/>
14. Alam N. Unmet need for additional medical care for sick children in mother's view in rural Bangladesh: implications for improving child health services. *J Biosoc Sci* [Internet]. 2007 Sep [cited 2025 Apr 7];39(5):769–78. Available from: <https://pubmed.ncbi.nlm.nih.gov/17316472/>
15. Naghavi M, Wang H, Lozano R, Davis A, Liang X, Zhou M, et al. Global, regional, and national age-sex specific all-cause and cause-specific mortality for 240 causes of death, 1990-2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet* [Internet]. 2015 Jan 10 [cited 2025 Apr 7];385(9963):117–71. Available from: <https://pubmed.ncbi.nlm.nih.gov/25530442/>
16. Labrique AB, Sikder SS, Wu L, Rashid M, Ali H, Ullah B, et al. Beyond pregnancy – the neglected burden of mortality in young women of reproductive age in Bangladesh: a prospective cohort study. *BJOG* [Internet]. 2013 Aug 1 [cited 2025 Apr 7];120(9):1085–9. Available from: <https://onlinelibrary.wiley.com/doi/full/10.1111/1471-0528.12245>

17. Hernández B, Ramírez-Villalobos D, Romero M, Gómez S, Atkinson C, Lozano R. Assessing quality of medical death certification: Concordance between gold standard diagnosis and underlying cause of death in selected Mexican hospitals. *Popul Health Metr* [Internet]. 2011 Aug 4 [cited 2025 Apr 7];9. Available from: <https://pubmed.ncbi.nlm.nih.gov/21816103/>
18. Naghavi M, Wang H, Lozano R, Davis A, Liang X, Zhou M, et al. Global, regional, and national age-sex specific all-cause and cause-specific mortality for 240 causes of death, 1990–2013: a systematic analysis for the Global Burden of Disease Study 2013. *Lancet* [Internet]. 2015 Jan 10 [cited 2025 Apr 7];385(9963):117–71. Available from: <https://pubmed.ncbi.nlm.nih.gov/25530442/>
19. Each drowning death is preventable: WHO and UNICEF [Internet]. [cited 2025 Apr 11]. Available from: <https://www.unicef.org/bangladesh/en/press-releases/each-drowning-death-preventable-who-and-unicef>
20. Hossain M, Mani KKC, Sidik SM, Hayati KS, Rahman AKMF. Socio-demographic, environmental and caring risk factors for childhood drowning deaths in Bangladesh. *BMC Pediatr* [Internet]. 2015 Sep 10 [cited 2025 Apr 11];15(1):1–6. Available from: <https://bmcpediatr.biomedcentral.com/articles/10.1186/s12887-015-0431-7>
21. Ahs JW, Tao W, Löfgren J, Forsberg BC. Diarrheal Diseases in Low-and Middle-Income Countries: Incidence, Prevention and Management. *Open Infect Dis J*. 2010;4:113–24.
22. Abbafati C, Abbas KM, Abbasi M, Abbasifard M, Abbasi-Kangevari M, Abbastabar H, et al. Global burden of 369 diseases and injuries in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *The Lancet* [Internet]. 2020 Oct 17 [cited 2025 Apr 11];396(10258):1204–22. Available from: <https://www.thelancet.com/action/showFullText?pii=S0140673620309259>
23. Naghavi M, Ong KL, Aali A, Ababneh HS, Abate YH, Abbafati C, et al. Global burden of 288 causes of death and life expectancy decomposition in 204 countries and territories and 811 subnational locations, 1990–2021: a systematic analysis for the Global Burden of Disease Study 2021. *The Lancet* [Internet]. 2024 May 18 [cited 2025 Apr 11];403(10440):2100–32. Available from: <https://www.thelancet.com/action/showFullText?pii=S0140673624003672>
24. Bryce J, Boschi-Pinto C, Shibuya K, Black RE. WHO estimates of the causes of death in children. *Lancet* [Internet]. 2005 Mar 26 [cited 2025 Apr 11];365(9465):1147–52. Available from: <https://www.thelancet.com/action/showFullText?pii=S0140673605718778>

25. Victora CG, Bryce J, Fontaine O, Monasch R. Reducing deaths from diarrhoea through oral rehydration therapy.
26. Blencowe H, Lawn J, Vandelaer J, Roper M, Cousens S. Tetanus toxoid immunization to reduce mortality from neonatal tetanus. *Int J Epidemiol* [Internet]. 2010 Apr 1 [cited 2025 Apr 8];39(Suppl 1):i102. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC2845866/>
27. Di Pietrantonj C, Rivetti A, Marchione P, Debalini MG, Demicheli V. Vaccines for measles, mumps, rubella, and varicella in children. *Cochrane Database Syst Rev* [Internet]. 2021 Nov 22 [cited 2025 Apr 8];2021(11):CD004407. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8607336/>
28. World Health Organization (WHO), Walke SP, Das R, Acharya AS, Pemde HK, Walke SP, et al. Acute Respiratory Infections in Children. *National Journal of Community Medicine* [Internet]. 2006 [cited 2025 Apr 8];5(2):15–20. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK11786/>
29. Esposito S, Principi N. Prevention of pertussis: An unresolved problem. *Hum Vaccin Immunother* [Internet]. 2018 Oct 3 [cited 2025 Apr 8];14(10):2452. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6284497/>
30. Nahar Q, Arifeen S El, Jamil K, Streatfield PK. Causes of adult female deaths in Bangladesh: Findings from two National Surveys Global health. *BMC Public Health* [Internet]. 2015 Sep 18 [cited 2025 Apr 16];15(1):1–9. Available from: <https://bmcpublichealth.biomedcentral.com/articles/10.1186/s12889-015-2256-6>

ORIGINALITY REPORT

8%

SIMILARITY INDEX

6%

INTERNET SOURCES

6%

PUBLICATIONS

0%

STUDENT PAPERS

PRIMARY SOURCES

1	www.globalhealthaction.net Internet Source	1%
2	Kemal Nazarudin Siregar, Rico Kurniawan, Dion Zein Nuridzin, Ryza Jazid Baharuddin Nur et al. "Strengthening causes of death identification through community-based verbal autopsy during the COVID-19 pandemic", BMC Public Health, 2022 Publication	1%
3	assets-eu.researchsquare.com Internet Source	<1%
4	hearing.health.mil Internet Source	<1%
5	microcon2018.com Internet Source	<1%
6	David Mulenga. "Understanding Public Health in Africa - Issues and Cases", Routledge, 2025 Publication	<1%
7	iris.paho.org Internet Source	<1%
8	link.springer.com Internet Source	<1%
9	reproductive-health-journal.biomedcentral.com Internet Source	<1%
10	Mosharaf Hossain, Kulanthayan K. C. Mani, Sherina Mohd Sidik, K. S. Hayati, A. K. M.	<1%

Fazlur Rahman. "Socio-demographic, environmental and caring risk factors for childhood drowning deaths in Bangladesh", BMC Pediatrics, 2015

Publication

-
- | | | |
|----|---|------|
| 11 | hubcymru.org.uk
Internet Source | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 12 | jogh.org
Internet Source | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 13 | www.ncbi.nlm.nih.gov
Internet Source | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 14 | www.thelancet.com
Internet Source | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 15 | www.who.int
Internet Source | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 16 | Baleng Mahama Wutor, Isaac Osei, Golam Sarwar, Williams Oluwatosin Adefila et al. "Under-5 mortality surveillance in low-income and middle-income countries: insights from two Health and Demographic Surveillance Systems in rural Gambia", BMJ Global Health, 2024
Publication | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 17 | Peter Serina, Ian Riley, Bernardo Hernandez, Abraham D. Flaxman et al. "The paradox of verbal autopsy in cause of death assignment: symptom question unreliability but predictive accuracy", Population Health Metrics, 2016
Publication | <1 % |
|----|---|------|
-
- | | | |
|----|---|------|
| 18 | Rohina Joshi, Neha Faruqui, Shilpa R Nagarajan, Rasika Rampatige, Alex Martiniuk, Hebe Gouda. "Reporting of ethics in peer-reviewed verbal autopsy studies: a systematic review", International Journal of Epidemiology, 2018 | <1 % |
|----|---|------|

19	apps.who.int Internet Source	<1 %
20	journals.lww.com Internet Source	<1 %
21	pure.umcutrecht.nl Internet Source	<1 %
22	uhra.herts.ac.uk Internet Source	<1 %
23	www.coursehero.com Internet Source	<1 %
24	www.frontiersin.org Internet Source	<1 %
25	www.pubmedcentral.nih.gov Internet Source	<1 %

Exclude quotes On

Exclude bibliography On

Exclude matches Off